Intoxicant Consumption Dynamics under Alcohol Prohibition: Evidence from India

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Abstract

This paper examines the effects of an alcohol prohibition law in Bihar, India, on intoxicant consumption. We implement a dynamic difference-in-difference estimation strategy using longitudinal data on monthly household expenses, exploiting state-level variation in policy exposure and household-level variation in alcohol use. We find that alcohol-consuming households in Bihar reduced tobacco expenditure alongside alcohol after the ban announcement, consistent with complementarity between alcohol and other intoxicants; however, over the following months of sustained alcohol unavailability, these households gradually increased tobacco expenditure, indicating a significant medium-run adjustment to tobacco after prohibition.

JEL Classification: D10, D12, H31, H75

Keywords: Alcohol prohibition, Tobacco expenditure, Household budgets, Dynamic difference-in-differences, Bihar

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1 Introduction

Policies that restrict addictive goods are often evaluated by their effects on the target substance, yet their broader consequences depend on how households reallocate spending across related intoxicants. Theory offers ambiguous predictions: if intoxicants are complements, restricting one reduces household spending on others; however, if they are substitutes, restrictions on one may shift consumption toward the alternatives available. This issue is particularly salient for alcohol, whose consumption has substantial health, social, and economic costs. Understanding how alcohol restrictions affect the consumption of other intoxicants–and whether these spillovers are temporary, persistent, or evolve over time–is therefore central to policy evaluation. The key question thus is not only whether such restrictions reduce alcohol use, but also whether they lead to shifts toward other addictive goods—and whether these responses become muted, reversed, or intensified over time.

We study this question using the 2016 alcohol prohibition in Bihar, India. The Bihar Amendment Act imposed a sweeping ban on the consumption, production, transportation and sale of alcohol in the state (Chakrabarti et al., 2024). We study this policy using high-frequency longitudinal data from the Consumer Pyramids Household Survey (CPHS) and examine how prohibition affected household expenditure on alcohol and on cigarettes, bidis, and other tobacco products (hereafter, CBT). Our main empirical finding is that the ban generated a dynamic and non-linear response in CBT expenditure. Alcohol spending declines sharply following the announcement and implementation of prohibition, as expected. CBT expenditure among exposed households also falls initially, but this decline later reverses, with tobacco spending recovering over the medium run. A specification focusing only on a single average post-treatment effect would therefore miss a key feature of the data: the cross-good response to prohibition is not monotonic.

Understanding the dynamic pattern of the cross-good response is the core contribution of the paper. In the short run, the evidence is consistent with a decline in joint consumption. However, in the medium run, households partially reallocate expenditure to tobacco as alcohol remains unavailable. More broadly, the paper shows that policy

evaluations based only on the targeted good or only on average post-treatment effects, can mischaracterise the wider behavioural consequences of restrictive intoxicant policies. The monthly structure of the Consumer Pyramids Household Survey (CPHS), combined with its rich and granular information on a wide range of household consumption items, is crucial for this analysis, as it allows us to examine the evolving relationship between alcohol and tobacco—specifically, whether the two goods behave as complements or substitutes—as households adjust following the policy change.

Our empirical strategy exploits the variation in exposure to the Bihar ban in two complementary ways. We begin with an interstate event-study design that compares households in Bihar with households in neighbouring states not subject to prohibition, focusing on differential changes between alcohol-consuming households. This comparison identifies the effect of the legal regime change while limiting concerns that broader national trends or distant regional shocks drive the results. We then complement this with a within-Bihar event-study that compares households with positive pre-ban alcohol consumption to households reporting no pre-ban alcohol consumption. This second comparison is useful because the policy applies statewide, but its direct exposure is concentrated among households that consumed alcohol prior to prohibition. Together, the two designs allow us to distinguish the effect of the ban from confounding changes in market conditions, societal perceptions, and local supply factors that could affect households in Bihar.

The timing of the policy is both unique and useful for identification. The ban was announced on November 26, 2015, and implemented on April 1, 2016, following sustained mobilisation by women and social activists who argued that alcohol consumption contributed to domestic abuse and general social harm. This sequencing allows us to separate anticipation effects from post-enforcement responses. In interstate analysis, alcohol expenditure in Bihar declines sharply relative to neighbouring states after the policy announcement and remains persistently lower thereafter, indicating a substantial contraction in recorded alcohol consumption. Over the same horizon, CBT expenditure also falls initially, but roughly a year after implementation, that decline begins to narrow.

The within-Bihar analysis yields a similar pattern: households that consumed alcohol before the ban reduce CBT expenditure relative to non-consuming households in the months following prohibition, but later exhibit a gradual recovery in tobacco spending. The consistency of these patterns across the two designs strengthens the interpretation that prohibition altered the time path of tobacco consumption among households most exposed to the loss of legal alcohol access.

We interpret the observed pattern as evidence that the relationship between alcohol and tobacco is dynamic rather than fixed. One possible explanation is that alcohol and CBT are complementary in the short run, so that an immediate reduction in alcohol use depresses tobacco consumption as well. Forward-looking households reduce alcohol consumption after the ban announcement, even before its implementation, because drinking now increases future addictive demand that cannot be satisfied once the ban begins. A later period of adjustment, approximately three months after the ban, households shift expenditure back towards tobacco as alcohol remains unavailable, with the gap widening between alcohol consuming and non-alcohol-consuming households over time. This behavior is consistent with the addiction syndrome model proposed by Shaffer et al. (2004), wherein households predisposed to some form of addiction substitute for other forms of addictive goods in the absence of the other. In addition to the interstate analysis, the within-state comparison is informative because it helps distinguish changes concentrated among previously alcohol-consuming households from shocks common to all households in Bihar, including complementarities between retail access to alcohol and tobacco or broader changes in the societal stigma associated with intoxicant consumption. The broader implication is that the effects of prohibition cannot be summarised by a single static treatment effect, nor can related intoxicants be cleanly classified as strict complements or substitutes without accounting for adjustment dynamics.

Our specifications control for a host of household-level observables to account for heterogeneity in intoxicant demand (Cotti et al., 2015), such as religion, family size, gender distribution, etc., and even those that vary over time, such as income. In addition, we subject these findings to a range of robustness checks. Since the CPHS follows the same

households over time, we find that migration across states does not drive our results. We also show that the results are not explained by price changes, are robust to excluding border districts, and remain similar if our standard errors are bootstrapped, and specifications based on expenditure shares. Across these exercises, the central qualitative finding is similar: the Bihar prohibition sharply reduced alcohol expenditure and induced a non-linear response in tobacco expenditure among exposed households.

Our findings are consistent with a broader literature showing that policies aimed at restricting the consumption of one addictive good can affect the consumption of other addictive goods. Existing work on tobacco regulation shows that increases in cigarette prices can induce substitution to alcohol (Picone et al., 2004; Decker and Schwartz, 2000) and smoking cessation products (Cotti et al., 2016), while reducing the use of e-cigarettes (Cotti et al., 2018).[1] The evidence on smoking bans is similarly heterogeneous, some studies finding increased alcohol consumption and others finding declines (Burton, 2025; Picone et al., 2004).[2] These studies suggest that policies targeting smoking can generate non-trivial spillovers across related intoxicant markets, with the direction of the response depending on the adjustment margin and the set of available alternatives.

Similarly, a related literature shows that alcohol-based policies also shape the consumption of other substances, although the evidence is mixed and highly context-dependent. Higher prices for specific types of alcohol can induce substitution for other forms of alcohol (Gehrsitz et al., 2021), while higher alcohol prices reduce smoking participation (Decker and Schwartz, 2000), consistent with complementarity between alcohol and tobacco in some settings. Shrestha (2015) argues that one's level of alcohol consumption is crucial while evaluating the effectiveness of using alcohol taxes. Spillover effects are not limited to alcohol and tobacco. Survey-based evidence further indicates that increases in the minimum legal drinking age can shift alcohol consumption to marijuana among young adults (DiNardo and Lemieux, 2001; Crost and Guerrero, 2012), while increasing the probability of marijuana use sharply at the age of 21 (Crost and Rees, 2013). Fur-

[1] There is also substantial work looking at the cigarette taxes (Hansen et al., 2017; Abouk et al., 2023; Dennett, 2024) and smoking bans (Carton et al., 2016) on smoking prevalence.

[2] Choi et al. (2019) investigates the impact of medical marijuana laws (MMLs) on tobacco cigarette consumption and finds a substantial reduction in smoking.

ther, late-night alcohol sales restrictions meant to impact adults can improve children's outcomes (Petrusevich, 2024). Further, Alpert et al. (2018), show that the introduction of an abuse-deterrent formulation of OxyContin led to increased heroin use. Therefore, the need is to study these complementarities and substitution effects to correctly evaluate the impact of such policies.[3] Taken together, this literature makes clear that policies targeting one addictive good can alter demand for other goods, but the sign and magnitude of these responses depend on the policy environment, the adjustment margin, and the population under study. Further, these studies primarily examine these policy changes in developed countries; evidence from developing countries remains limited.

Our paper makes three contributions. First, we study the effects of a comprehensive alcohol prohibition, rather than marginal policy changes such as taxes, sales restrictions, or age limits that dominate the literature. A complete ban represents a discrete shift in legal access and can induce behavioural responses that differ fundamentally from those generated by incremental interventions. We provide evidence on how such a prohibition affects tobacco use, complementing prior work that examines the impact of the Bihar alcohol ban on crime (Dar and Sahay, 2018; Chaudhuri et al., 2024), on fertility and on children's health (Dar and Vellakkal, 2026) and learning outcomes (Dar et al., 2026). We are also able to distinguish between the impact of policy announcement and policy implementation.

Second, we show that the cross-good relationship between alcohol and tobacco is dynamic and exploit a unique empirical setting that combines a sharp prohibition-induced shock with high-frequency household-level data to trace this adjustment precisely. The granularity of our data allows us to precisely trace this time horizon and identify how the cross-good relationship unfolds over time. Tobacco expenditure declines immediately after prohibition but subsequently recovers, implying that both the sign and magnitude of the response depend on the time horizon. While most existing studies estimate static relationships, our setting allows us to cleanly observe short- and medium-run responses to a large, discrete change in alcohol availability, demonstrating that substitution pat-

[3]Other studies on opioid and marijuana consumption, Mallatt (2022), Anderson and Rees (2013), Betz and Jones (2022), Anderson et al. (2018)

terns evolve over time. This complements evidence such as Harris et al. (2024), which documents lifecycle variation in the relationship between cannabis and tobacco, by highlighting temporal adjustment following an abrupt policy shock.

Third, we provide evidence from a developing-country context, where both the constraints faced by households and the impacts of intoxicant use differ markedly from high-income settings. Lower incomes, limited access to healthcare, and weaker enforcement capacity can amplify the welfare consequences of substance use and shape behavioural responses to prohibition. Despite this, existing evidence from developing countries largely relies on price variation (Chandra and Doshi, 2025; Lee, 2007; Yu and Abler, 2010) and recovers static cross-good relationships (Rahman, 2003; John, 2006, 2008). Our findings highlight how responses to large, discrete policy shocks may differ in such settings, underscoring the importance of context for policy design and the need to evaluate its evolving relationship over time. Falézan (2020); Dar and Vellakkal (2026) documents a post-ban decline in tobacco expenditure, consistent with alcohol-tobacco complementarity. However, using our disaggregated monthly event-study estimates, we show that this effect is temporary and the relationship reverses over time, a pattern supported by both our interstate and within-Bihar analyses, suggestive of a cross-good addictive demand response.

The remainder of the paper is structured as follows. Section 2 provides background on the Bihar liquor ban. Section 3 describes the data and Section 4 describes the empirical strategy. Section 5 presents the main results, Section 6 discusses robustness checks, and Section 7 concludes.

2 Background and Conceptual Framework

India's federal structure grants individual states substantial autonomy over alcohol policy, including the minimum legal drinking age, sales regulation, and excise taxes.[4] Bihar, the third most populous Indian state according to the 2011 Census, ranked sixth nationally in alcohol consumption in the 2011–2012 National Sample Survey (Chaudhuri et al., 2024).

[4]See Schess et al. (2023) for a comprehensive review of alcohol policies in India.

Bihar is also one of only two Indian states to have enacted a nearly complete liquor prohibition in recent years.[5]

During his tenure as Chief Minister, Nitish Kumar liberalised alcohol retailing, contributing to a substantial increase in liquor consumption in Bihar. As shown in Figure 1a, pre-ban liquor expenditure in Bihar exceeded both the national average and the neighbouring states. In July 2015, ahead of the state elections and in response to sustained protests by women's groups and social activists, Kumar pledged to implement prohibition if re-elected. These groups argued that alcohol consumption was a major driver of domestic abuse and intimate partner violence. On 26 November 2015, shortly after his re-election, Kumar announced that Bihar would enact a comprehensive alcohol ban effective from April 2016. The Bihar Excise (Amendment) Bill 2016, which was passed unanimously, prohibited the production, sale, and consumption of all forms of alcohol in the state (Singh, 2020). Violations were subject to severe penalties, including imprisonment of up to ten years. Enforcement was also stringent: the government publicised a toll-free hotline for reporting violations, and police and excise authorities carried out coordinated raids and arrests (Chaudhuri et al., 2024).[6]

Figure 1a plots the average monthly household alcohol expenditure in Bihar, its neighbouring states (Uttar Pradesh, West Bengal, and Jharkhand) and the rest of India excluding Bihar between 2015 and 2017.[7] Prior to the ban, the average monthly household alcohol expenditurein Bihar exceeded the national average by roughly 75 INR and that of neighbouring states by around 100 INR. Alcohol expenditure in Bihar began to fall about three months before the formal announcement, coinciding with the election campaign during which prohibition was first promised. While average alcohol expenditure elsewhere in India remained broadly stable, Bihar experienced a sharp decline between the policy announcement and its implementation. This pattern is at odds with the simple

[5]Gujarat has enforced a statewide ban since the 1950s, but with important exceptions for visitors and continued legal sales in the neighbouring Union Territory of Diu. Intermittent and less strictly enforced bans have also existed in Nagaland, Manipur, and Mizoram.

[6]Between April 2016 and March 2018, authorities conducted more than 670,000 raids and made 126,000 arrests for alcohol-related offences (Excise and Prohibition Department, Government of Bihar).

[7]These neighbouring states share a border with Bihar and have broadly comparable social and economic characteristics; Jharkhand was part of Bihar until 2001.

stockpiling hypothesis and is more consistent with tightening supply conditions in anticipation of prohibition. There is also some suggestive evidence of spillovers to neighbouring states, which experienced a modest decline in liquor expenditure, while the rest of India remained largely unchanged. Following implementation, average household alcohol expenditure in Bihar declined further and eventually fell to zero.

Figure 1b presents analogous trends for spending on cigarettes, bidis, and other tobacco products (CBT). At the start of 2015, average CBT expenditure in Bihar was roughly 100 INR below that of neighbouring states and broadly similar to the national average. In the six months between the announcement and the implementation of the ban, Bihar experienced a sharp decline in CBT expenditure, mirroring the fall in alcohol spending. However, this decline did not persist. Within approximately 3 months of implementation, CBT expenditure began to recover, reversing much of the initial fall. During the same period, the average CBT expenditure rose modestly in the rest of India and declined only slightly in the neighbouring states. These aggregate patterns foreshadow the central result of the paper: the response of tobacco expenditure to alcohol prohibition is dynamic rather than constant over time. They also motivate the subsequent household-level analysis.

2.1 Theoretical Insight

To understand the dynamic response to a ban, let us consider a simple model with three periods: a pre-announcement period, an announcement period before enforcement, and a post-ban period. Households choose to allocate their resources between alcohol, tobacco and other forms of consumption in each period. Alcohol and tobacco are both addictive goods; therefore, their utility from these goods depends on their stock of past consumption of these intoxicants, similar to the rational addiction model proposed by Becker and Murphy (1988). If alcohol and tobacco are assumed to be complements, marginal utility for one good increases with an increase in consumption of the other; therefore, any change in one will lead to a similar change in the other. At the same time, current consumption contributes to a stock of addictive demand that carries into the future. Once the ban

is announced, rational forward-looking households would recognize that future alcohol consumption will be impossible after implementation (period 3). This lowers the value of consuming alcohol in the announcement period (period 2), since current drinking raises future addictive demand that cannot later be satisfied with alcohol. Because tobacco and alcohol are complements, tobacco consumption would also be expected to fall initially. After implementation, however, households still carry a positive addictive-demand stock into the post-ban period. With alcohol unavailable, tobacco becomes the remaining intoxicant margin. According to the addiction syndrome model proposed by Shaffer et al. (2004), it would generate substitution toward tobacco among those predisposed to alcohol use and cause a recovery in tobacco consumption.

3 Data

We use monthly longitudinal household data from the Consumer Pyramids Household Survey (CPHS), conducted by the Centre for Monitoring Indian Economy Pvt. Ltd. (CMIE). The CPHS follows more than 170,000 households over time and is conducted in three four-month waves each year: January–April, May–August, and September–December. Each household is assigned a specific interview month within a wave and is then re-interviewed exactly four months later. Monthly household time series on income and expenditure are constructed from information collected at each interview on the four months preceding the interview month. Table 1 provides the summary statistics.

The CPHS has several features that make it particularly well-suited for our analysis. First, it is the only dataset that consistently surveys a representative set of households before, during, and after the imposition of the Bihar ban, allowing us to compare Bihar with neighbouring states as well as conduct within-state analysis between households that reported alcohol consumption at baseline and those that did not. Second, the survey contains a highly disaggregated breakdown of household expenditure across 153 categories, with additional detail for many items. These include food and intoxicants, clothing, footwear, cosmetics, appliances, restaurants, utilities, transport, education, and health.

The data also provide detailed information on household income and its sources, which allows us to control for income fluctuations in the regressions.[8] Third, and most importantly for our purposes, the high-frequency structure of the data allows us to estimate dynamic difference-in-differences specifications that distinguish between responses to the policy announcement, the period between announcement and implementation, and the months following enforcement. This makes it possible to study adjustment dynamics that would be obscured in standard average-treatment-effect estimates.

Because the ban was announced in November 2015 and implemented in April 2016, we restrict the sample to the period from January 2015 to December 2017. This window allows us to observe household expenditure up to ten months before the announcement of prohibition, the response to the announcement itself, and both the immediate and medium-run effects of implementation. Although the CPHS would allow us to extend the analysis further, we do not do so for two reasons. First, communal riots in Bihar in March 2018 led to the destruction of shops and public and private property. Second, major fluctuations in tobacco prices occurred in Bihar between August and October 2018, potentially as changes in the tax structure took effect. Restricting the sample to December 2017 helps isolate the effects of the alcohol ban from these later shocks. We focus on detailed expenditure in the categories of alcohol and intoxicants, including cigarettes, bidis, and other tobacco products (CBT).[9]

4 Empirical Strategy

We estimate the causal impact of the Bihar alcohol ban using a dynamic difference-in-differences framework that compares the spending patterns of treated and control households over time. This approach is well suited to the policy setting, because prohibition was first announced and only implemented six months later, allowing household behaviour

[8] There are two other important sources of household consumption data for India. The long-running National Sample Survey is cross-sectional, and the most recent publicly available comprehensive round is from 2011–12; micro-data from the 2022–23 round are not yet publicly available. The Indian Human Development Survey (IHDS), while longitudinal, is available only for 2004–05 and 2011–12.

[9] We winsorise expenditure variables at the 1st and 99th percentiles to reduce the influence of outliers.

to adjust both in anticipation of and in response to the enforcement.[10] To separate the effects of prohibition from other policy, secular, and market trends, we exploit the timing of the ban in two complementary settings: an interstate comparison and a within-Bihar comparison. We focus on two expenditure outcomes: (i) alcohol and (ii) cigarettes, bidis, and other tobacco products (CBT). Our main specification is:

$$\begin{aligned} Y_{it} =& \beta_0 + \beta_1 \text{Treated}_i + \sum_{k=-2}^{-10} \gamma_{\text{pre},k} \cdot 1[D_t = -k] \times \text{Treated}_i \\ &+ \sum_{k=0}^{20} \gamma_{\text{post},k} \cdot 1[D_t = k] \times \text{Treated}_i + \beta_4 X_i + \beta_5 X_{it} + \alpha_d + \alpha_t + \epsilon_{it}, \end{aligned} \tag{1}$$

where Y_{it} denotes expenditure on $Y \in \{\text{Alcohol, CBT}\}$ by household i in month t.

The indicator Treated_i is equal to one for households in Bihar that reported positive alcohol expenditure between January 2015 and October 2015, that is, in the ten months prior to the announcement of prohibition. These are the households most directly exposed to the ban. Keeping this treatment group fixed, we estimate the specification using two alternative control groups. In the first exercise, we implement an interstate comparison in which the control group consists of households with positive pre-ban alcohol expenditure in the neighbouring states of Bihar, namely West Bengal, Jharkhand and Uttar Pradesh. In the second exercise, we perform a comparison within-Bihar in which the control group consists of households in Bihar with zero alcohol expenditure between January and October 2015. The interstate design identifies the effect of the policy relative to nearby states not subject to prohibition, while the within-state design helps address concerns that evolving state-level conditions or cross-border spillovers may contaminate the interstate comparison.

The specification includes district fixed effects, α_d, to absorb time-invariant cross-sectional differences across districts. This is particularly useful in the present context, as it reduces the risk that estimates are driven by persistent differences in districts where

[10] We also estimate average treatment effects using two indicator variables, one for the period between announcement and implementation and another for the post-implementation period. The specification and results are reported in Appendix A.1. However, these average effects mask the dynamic adjustment in intoxicant consumption, especially for CBT, and are therefore less informative than the event-study estimates presented in Section 5.

alcohol may have remained more readily available through informal channels. Time fixed effects, α_t, absorb aggregate shocks common to all households in a given specification. In interstate comparison, standard errors are clustered at the state level. In the within-Bihar comparison, the benchmark specification clusters standard errors at the district level.[11] We also report wild-cluster bootstrap results to address concerns about the small size of the clusters.

Our coefficients of interest are the event-study estimates $\gamma_{\text{post},k}$, which are presented graphically in Section 5. The coefficient for October 2015, the month immediately preceding the announcement, is normalised to zero. The post-treatment coefficients therefore measure, month by month, the differential evolution of expenditure for treated households relative to the relevant control group after the announcement of prohibition. The coefficients $\gamma_{\text{pre},k}$ capture the relative differences in expenditure between the treated and control households in the months prior to the announcement. These pre-treatment coefficients allow us to assess the plausibility of parallel trends, while the post-treatment coefficients trace the dynamic evolution of the treatment effect over time. The identifying assumption is that, absent prohibition, the difference in expenditure between treated and control households would have evolved similarly over time.

We also control for a set of household characteristics that are either time-invariant or time-varying. The vector X_i includes fixed household characteristics such as religion, caste, and region, which are relevant because the norms surrounding alcohol consumption differ systematically between social groups. Alcohol consumption patterns also vary between urban and rural areas. The vector X_{it} includes time-varying controls such as household size, total household income, and household education level.[12] Controlling for income is particularly important because our outcomes are measured in expenditure terms, and because changes in alcohol access may affect the productivity and earnings of household members, thereby influencing CBT expenditure through an income channel (Johansson et al., 2014). As discussed later, we also explore the effect of the ban on food

[11] Bihar has 38 districts.

[12] We use the CPHS household education-group variable, which is a categorical measure constructed from the education levels of household members.

and clothing expenditure to study the income effect induced by potential supply-side factors.

Potential threats to identification. A first concern is that post-ban alcohol expenditure may be under-reported because prohibition made alcohol consumption legally risky. Several features of the setting mitigate this concern. Most importantly, the sharp decline in reported alcohol expenditure begins in the months following the policy announcement and before the full implementation of the ban. This timing is more consistent with a real contraction in alcohol consumption, potentially driven by anticipatory supply-side responses, than with purely strategic misreporting after enforcement.

A second concern, common in studies of prohibition, is that some households may substitute illicit alcohol through black-market purchases, cross-border sourcing, or home brewing (Kumar and Prakash, 2016; Aggarwal et al., 2025). Although such channels may have limited the extent of the decline, a growing body of evidence suggests that the Bihar ban nevertheless reduced alcohol use in a substantively meaningful way. Agarwal et al. (2017), for example, document a 64 percent decline in alcohol use based on interviews with relatives of drinkers, together with improvements in the allocation of household resources. Other work finds reductions in household alcohol expenditure and improvements in children's educational outcomes (Dar et al., 2026), increases in nutritional intake (Krishnatri and Vellakkal, 2024), and declines in intimate partner violence (Dixit et al., 2023; Debnath et al., 2023) and crime (Chaudhuri et al., 2024). Taken together, this evidence suggests that even if some illicit consumption persisted, the ban substantially reduced alcohol use and generated broader social effects.

Consequently, our analysis proceeds under the assumption that prohibition produced a substantial, though not necessarily complete, reduction in alcohol consumption in Bihar. This is sufficient for our purposes, since the object of interest is the downstream response of CBT expenditure to a sharp decline in alcohol access and use. Importantly, there is no comparable policy incentive for households to under-report CBT expenditure, which strengthens confidence in our main outcome measure.

A final concern is that spillovers across state borders may violate the Stable Unit

Treatment Value Assumption (SUTVA) in interstate comparison. To address this issue, we complement the interstate design with the within-Bihar comparison between pre-ban alcohol-consuming and non-consuming households. Because all households in this exercise are located within the same state policy environment, within-state estimates are less susceptible to cross-border spillovers and provide an additional source of evidence on the dynamic relationship between alcohol and other intoxicants.

5 Results

We begin by estimating the effect of the Bihar alcohol ban on alcohol expenditure and then turn to its effects on expenditure on cigarettes, bidis, and other tobacco products (CBT). Throughout, we use the event-study specification in equation (1) and plot the estimated coefficients $\hat{\gamma}_{pre,k}$ and $\hat{\gamma}_{post,k}$ for the relevant treatment-control comparisons. Our focus is not only on whether prohibition changed expenditure, but also on how those effects evolved over time.

5.1 Bihar vs. neighbouring states

Figure 2a presents the estimated differences in monthly household alcohol expenditure from the interstate comparison. Following the announcement of prohibition, alcohol-consuming households in Bihar reduce their alcohol expenditure sharply relative to comparable households in neighbouring states. By the months immediately preceding implementation, alcohol expenditure for the treated group is lower by roughly 100 INR relative to the control group, compared to the baseline month of October 2015. After the ban comes into force in April 2016, alcohol expenditure in Bihar falls further and remains persistently below that of neighbouring states throughout the event-study window. This pattern is consistent with a substantial and sustained contraction in recorded alcohol consumption after prohibition.

Next, we examine whether this reduction in alcohol expenditure is accompanied by changes in expenditure on tobacco products. Figure 2b plots the corresponding event-

study coefficients for CBT expenditure in the interstate comparison. Before the announcement of prohibition, the estimated differences between Bihar and neighbouring states are relatively small and, for the three months immediately preceding the announcement, statistically indistinguishable from zero. Following the announcement, households in Bihar gradually reduce CBT expenditure relative to comparable households in neighbouring states, the differential falling by approximately 50 INR relative to the baseline by the time of implementation. This lower level of CBT expenditure persists for a few months. However, thereafter the pattern changes: CBT expenditure in Bihar begins to recover relative to neighbouring states, and by the end of the event-study window, the initial decline has largely reversed.[13]

This non-monotonic path is central to the interpretation of the policy. A static difference-in-differences estimate would compress these dynamics into a single average effect and would therefore miss a key feature of the response to prohibition. The event-study results instead show that the cross-good response evolves over time: the ban is associated first with a decline in CBT expenditure and later with a partial recovery. Several mechanisms could explain this pattern. One possibility is a short-term complementarity between alcohol and CBT consumption, whereby the initial fall in alcohol use reduces CBT expenditure as well, followed by a later adjustment towards tobacco as alcohol remains unavailable. As discussed earlier, this behavior is consistent with the addiction syndrome model proposed by Shaffer et al. (2004). Other mechanisms, such as complementarities in retail access or changes in the broader social perception of addictive goods, may also play a role. The interstate evidence alone cannot fully distinguish among these channels, which motivates the within-Bihar analysis that follows. For completeness, Appendix-A.1 reports standard average treatment effect estimates for the post-announcement and post-implementation periods; these estimates are informative about average changes in expenditure, but they necessarily compress the underlying dynamics and therefore miss the reversal in CBT expenditure that emerges in the event-study graphs.

[13] In Tables 2 and 5, we report the wild-clustered bootstrap estimates and p-values.

5.2 Within Bihar comparisons

Although the interstate comparison is informative, it cannot fully separate the effect of prohibition from other Bihar-specific shocks occurring at the same time, such as changes in local market conditions, supply networks, or state-level attitudes towards intoxicant consumption. We therefore complement the interstate analysis with a within-Bihar comparison that relies on differential exposure to the policy at the household level. Specifically, we define the treatment group as households that reported positive alcohol expenditure in the ten months prior to the announcement of prohibition, and the control group as households that reported zero alcohol expenditure over the same period.[14] This comparison allows us to examine whether households that previously consumed alcohol adjusted their CBT expenditure differently from those that did not consume alcohol prior to the policy.[15]

Figure 3a presents the corresponding estimates for alcohol expenditure. In the baseline month of October 2015, treated households spent, on average, about 215 INR more on alcohol than control households, which is mechanically consistent with the way the two groups are defined. Relative to this baseline, no pre-trends are observed. However, after the announcement of prohibition in November 2015, the alcohol expenditure gap narrows rapidly, falling by roughly 177 INR over the following five months, even before implementation. Once the ban is in place, the differential decreases further and is effectively eliminated. This pattern confirms that the policy sharply reduced alcohol expenditure among the households most directly exposed to prohibition.

Figure 3b shows the corresponding results for CBT expenditure. Before the announcement, treated households spend around 58 INR more on CBT than control households, indicating that pre-ban alcohol consumers also consumed more tobacco products on average. However, compared to the baseline month of October 2015, the pre-announcement estimates are not statistically different from each other. Following the announcement of prohibition, this CBT expenditure gap narrows substantially. Over the subsequent year,

[14] Average expenditure on alcohol and CBT over time for treated and control households in Bihar is shown in Figure 9 in the Appendix.

[15] Anger et al. (2011) shows that smoking bans can have heterogeneous effects on household behaviour depending on exposure to the restriction.

the estimated difference falls by roughly 100 INR relative to the baseline, indicating that households that had previously consumed alcohol reduced CBT expenditure by more than non-consuming households. However, about a year after the ban implementation, the pattern begins to reverse: the expenditure gap starts to widen again, and by the end of the sample, treated households have increased CBT expenditure relative to the control group by around 50 INR compared with the trough of the response. In other words, the within-Bihar comparison reproduces the same qualitative pattern as the interstate analysis: an initial decline in CBT expenditure followed by a later recovery.

The within-state analysis is also informative about interpretation. Because both treated and control households are located in Bihar, they are exposed to the same statewide policy environment, enforcement regime, and broader changes in social attitudes towards intoxicants. This means that the differential change in CBT expenditure between the two groups is less likely to be driven solely by state-level supply shifts or common changes in social disapproval. The larger initial decline in CBT expenditure among pre-ban alcohol-consuming households is consistent with a complementarity channel linking alcohol and tobacco consumption. At the same time, the later increase in CBT expenditure among these same households suggests a medium-term adjustment towards other intoxicants when alcohol remains unavailable. The evidence, therefore, points to a dynamic relationship between alcohol and CBT: the short-run response is consistent with complementarity, while the medium-run response indicates partial substitution or behavioural reallocation among households with greater pre-existing demand for intoxicants.[16].

6 Robustness Checks

We assess the robustness of our baseline estimates through a series of additional exercises. A first concern is that, given the severity of the penalties and the comprehensive

[16] An alternative explanation of this behavior could include greater access to illegal alcohol over time, which, due to its complementarity with tobacco, leads to an increase in CBT expenditure. However, as discussed in Section 4, evidence from the literature that has evaluated the effectiveness of the Bihar ban suggests that the ban was largely successful in reducing alcohol consumption for households. Therefore, we do not expect this to be the dominant channel explaining the results.

nature of the prohibition, some households may have migrated out of Bihar after the ban. The ban may also have affected migration indirectly through labour-market adjustments associated with the closure of legal alcohol production and distribution. To address these concerns, we first restrict the sample to households in Bihar that are observed throughout the study period.[17] As shown in Figure 4, alcohol and tobacco consumption follow a similar pattern to the benchmark results; the standard errors are larger, which presumably is driven by the smaller sample size. We also find a very similar result in the top panel of Figure 4 where we analyse a balanced panel for our interstate analysis. Next, we replace expenditure levels with expenditure shares, defined as the share of total household expenditure devoted to alcohol and CBT, in order to account for the possibility that households re-optimise their consumption bundles following prohibition. The resulting estimates, reported in Figure 5, remain consistent with the baseline findings.

Next, as a placebo test, we check the effect of the ban on food (excluding intoxicants) and clothing expenditure for alcohol consuming households in Bihar relative to the neighbouring states and also between alcohol consuming and non-alcohol consuming households within Bihar. Figure 6 shows that the alcohol ban did not lead to any significant changes in clothing expenditure for liquor-consuming households in Bihar relative to its neighbouring states. While prior to the ban, there were no significant differences in food spending across these two types of households, food expenses increased for those in Bihar relative to the others after the ban announcement. This coincides with their decrease in liquor and tobacco consumption, thereby indicating a reallocation among consumption bundles; this gap, however, reduces as households increase their tobacco consumption in the medium run. The difference in expenditures in food and clothing for within Bihar households also remains the same through our sample period. This exercise illustrates the distinct dynamics of liquor and tobacco (relative to other forms of household expenditure) in response to the prohibition policy, compared with other forms of household expenditure, and adds evidence that this behavior is not driven by the income effect induced by liquor business closures.

[17]Selective migration from Bihar may also contribute to the widening of confidence intervals in the interstate comparison as the event-study horizon lengthens.

A further concern is that households in border districts may continue to access alcohol from neighbouring states, which could attenuate the estimated effects of prohibition. To address this possibility, we exclude from the within-Bihar analysis all districts that share a border with another state. This restriction yields a sample of households less likely to access alcohol from outside Bihar. As shown in Figure 7, the resulting estimates are very similar to the benchmark results.

Finally, we examine whether changes in CBT prices could confound the expenditure patterns we document. Using a simple difference-in-differences exercise, we show that CBT prices in Bihar do not display unusual shifts or volatility during the event-study window relative to the comparison groups. Figure 8 therefore suggests that the dynamic response of CBT expenditure is unlikely to be driven by contemporaneous price movements.

7 Conclusion

This paper studies how households adjust their expenditure on other intoxicants when legal access to alcohol is abruptly removed, leveraging a comprehensive prohibition in a developing-country setting and high-frequency household data to provide new evidence on how cross-good responses evolve over time. We show that while prohibition sharply reduces alcohol expenditure, its effects on other intoxicants are dynamic. Expenditure on cigarettes, bidis, and other tobacco products (CBT) declines immediately following the announcement and implementation of prohibition, but subsequently recovers over time.

The within-Bihar analysis helps clarify this pattern. Relative to non-consuming households, pre-ban alcohol-consuming households reduce both alcohol and CBT expenditure in the short run, but later increase CBT spending as the ban persists. This evolution points to a shifting cross-good relationship rather than a fixed one: in the short term, the evidence suggests complementarity, with lower alcohol use accompanied by lower tobacco consumption, while over time households partially substitute toward tobacco as alcohol remains unavailable. Although broader changes in supply conditions or social norms

may also play a role, the within-state comparison highlights the central importance of differential exposure to the loss of alcohol access.

Taken together, these findings underscore the importance of adopting a dynamic perspective when evaluating prohibition policies. A static treatment-effect approach would miss a key feature of the response: the impact on related intoxicants changes over time. As a result, evaluations that focus only on the targeted good or on average post-treatment effects may understate or mischaracterize the broader behavioural adjustments induced by prohibition. More generally, our results suggest that restrictive intoxicant policies can generate unintended and evolving responses across related goods, with implications for both policy design and welfare assessment.

8 Acknowledgements

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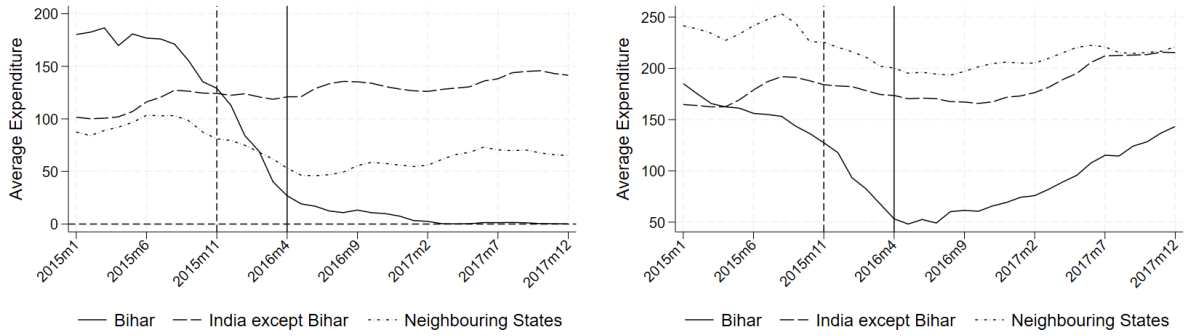
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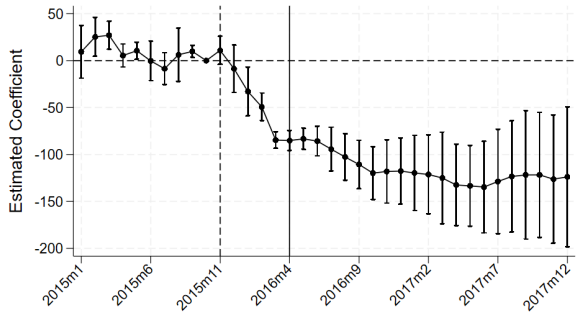


(a) Liquor Expenses

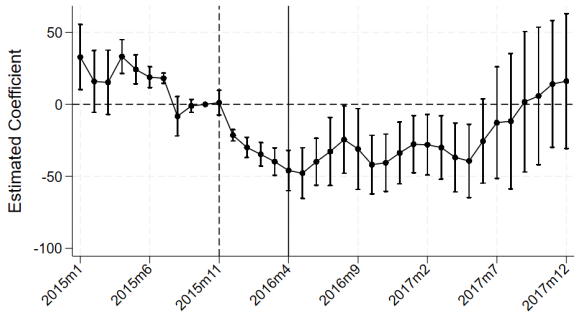
(b) CBT Expenses

Notes: This series comes from the Consumer Pyramids Household Survey (CPHS). Panel (a) plots the average monthly expenses for liquor in the Indian state of Bihar, its neighbouring states, including West Bengal, Jharkhand, and Uttar Pradesh, and all Indian states except Bihar (ROI); panel (b) plots the average monthly expenses for cigarettes, bidis, and other tobacco (CBT) in Bihar, its neighbouring states, and the rest of India. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

Figure 1: Average Monthly Expenditure on Liquor and CBT



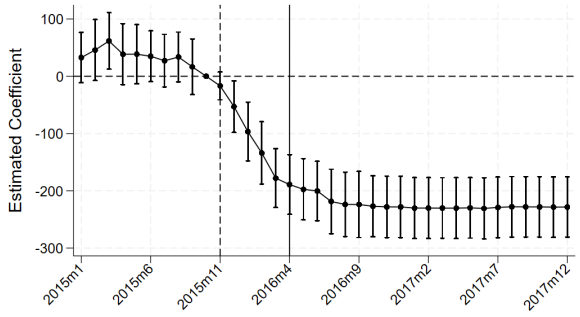
(a) Alcohol Consumption



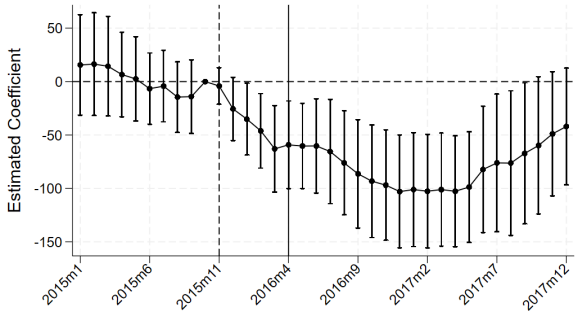
(b) Cigarettes, Bidis, Tobacco

Notes: Each graph includes point estimates from the event study exercise (normalised to 0 in October 2015) comparing alcohol-consuming households in Bihar with the neighbouring states. Regressions include district-fixed effects and time (month × year) fixed effects and are weighted to be representative at the state level. The confidence intervals are at the 95 percent level and are adjusted for state-level clustering. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

Figure 2: Interstate Event Study Results for Intoxicant Consumption



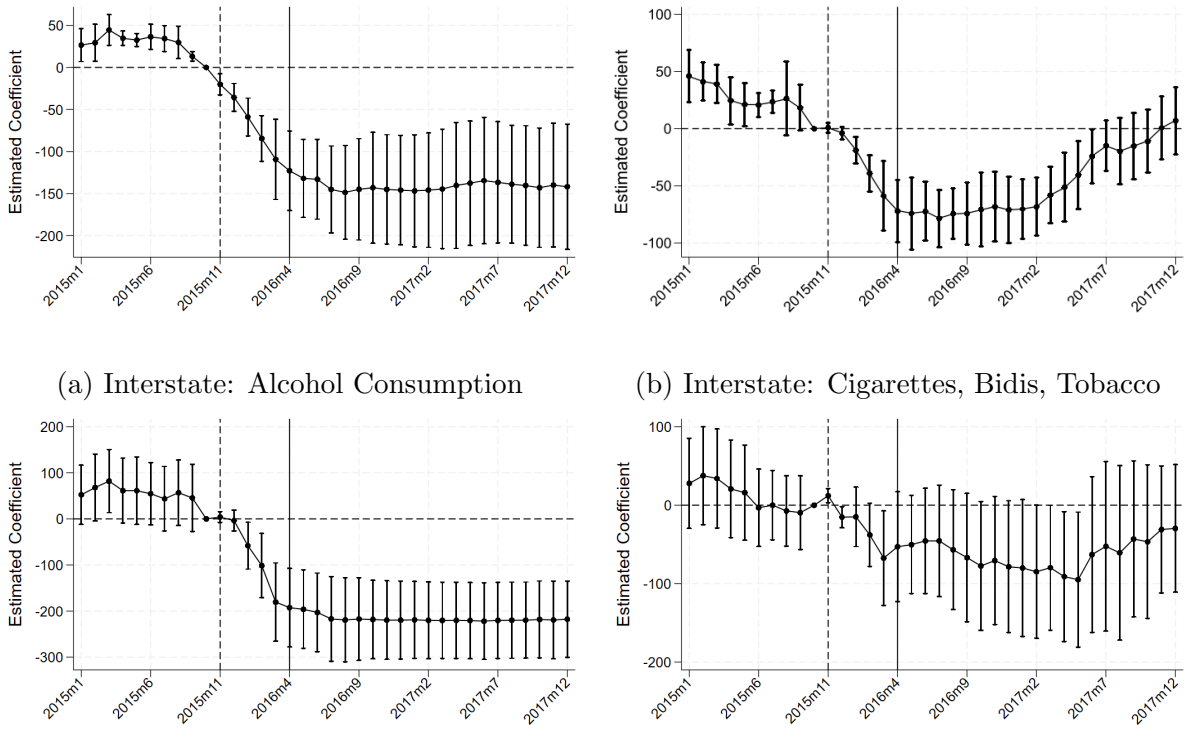
(a) Alcohol Consumption



(b) Cigarettes, Bidis, Tobacco

Notes: Each graph includes point estimates from the event study exercise (normalized to 0 in October 2015) comparing alcohol-consuming and non-alcohol-consuming households in Bihar. Regressions include district-fixed effects and time (month × year) fixed effects and are weighted. The confidence intervals are at the 95 percent level and are adjusted for district-level clustering. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

Figure 3: Event Study Results for Intoxicant Consumption within Bihar



(a) Interstate: Alcohol Consumption

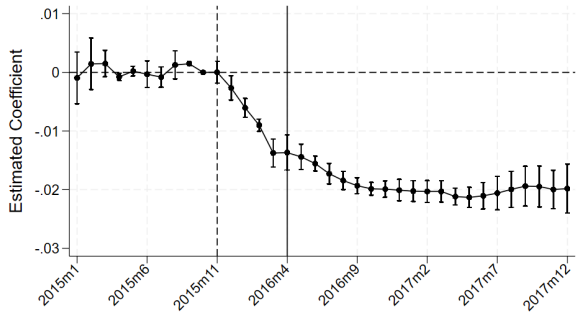
(b) Interstate: Cigarettes, Bidis, Tobacco

(c) Within Bihar: Alcohol Consumption

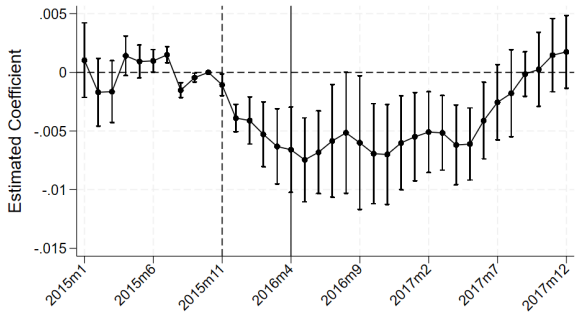
(d) Within Bihar: Cigarettes, Bidis, Tobacco

Notes: We use a balanced panel to control for the changing composition of households in our sample due to policy-induced migration. Each graph includes point estimates from the event study exercise (normalised to 0 in October 2015). Panels (a) and (b) show the interstate comparisons between alcohol-consuming households, whereas Panels (c) and (d) show the comparisons between alcohol-consuming and non-alcohol-consuming households in Bihar. Regressions include district fixed effects and time (month × year) fixed effects and are weighted. Confidence intervals are at the 95 percent level and adjusted for state-level clustering in the Interstate comparison and at the district-level clustering in the within-Bihar comparison. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

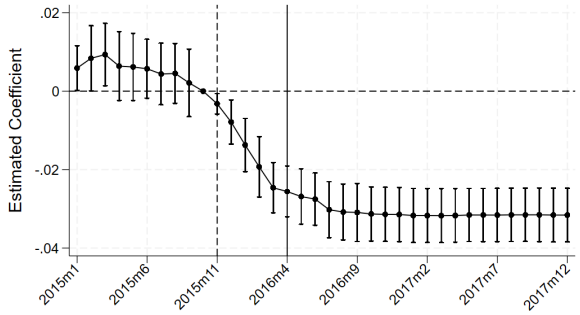
Figure 4: Event Study Results for Intoxicant Consumption: Balanced Panel



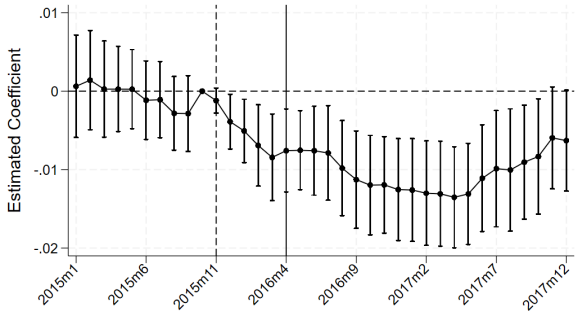
(a) Interstate: Alcohol Consumption



(b) Interstate: Cigarettes, Bidis, Tobacco



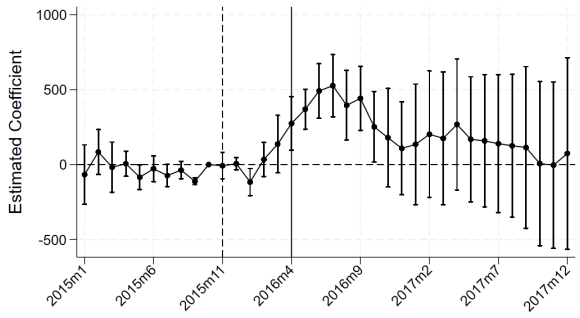
(c) Within Bihar: Alcohol Consumption



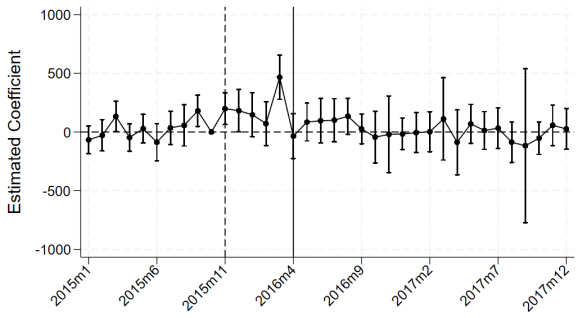
(d) Within Bihar: Cigarettes, Bidis, Tobacco

Notes: To understand how these households optimize and reallocate their resources, we investigate consumption shares instead of monthly expenses. Each graph includes point estimates from an event study exercise (normalised to 0 in October 2015). Panels (a) and (b) show the interstate comparisons between alcohol-consuming households, whereas Panels (c) and (d) show the comparisons between alcohol-consuming and non-alcohol-consuming households in Bihar. Regressions include district fixed effects and time (month × year) fixed effects and are weighted to be representative at the state level. Confidence intervals are at the 95 percent level;the interstate comparisons are adjusted for state-level clustering, and the within-Bihar comparisons are adjusted for district-level clustering. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

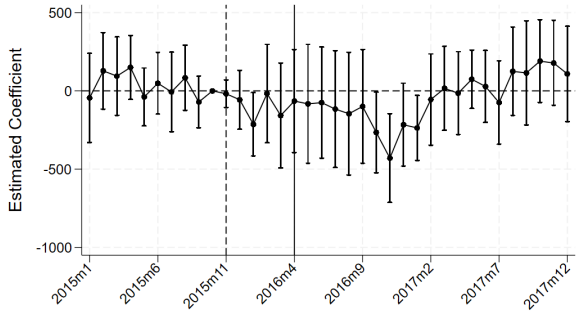
Figure 5: Event Study Results for Intoxicant Consumption: Consumption Shares



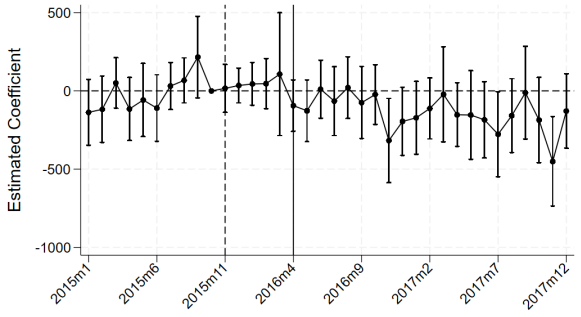
(a) Interstate: Food expenditure



(b) Interstate: Clothing Expenditure



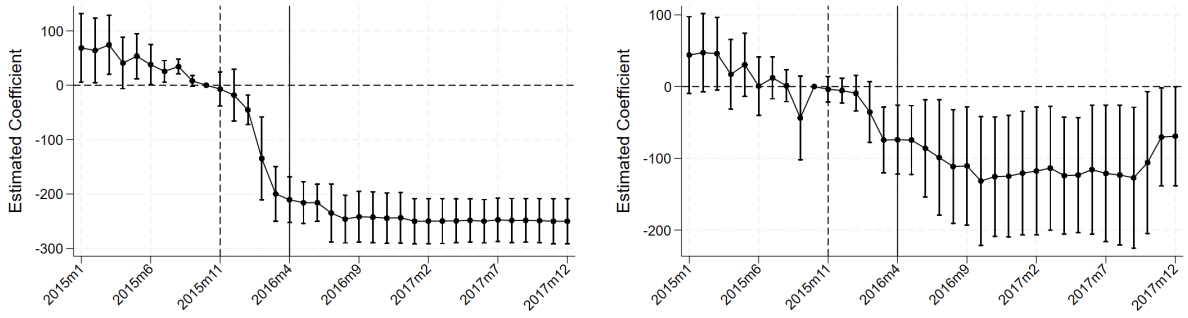
(c) Within Bihar: Food expenditure



(d) Within Bihar: Clothing Expenditure

Notes: Each graph includes point estimates from the event study exercise (normalised to 0 in October 2015) comparing food and clothing expenses between alcohol-consuming households in Bihar and in the neighbouring states in panel (a) and (b) and between alcohol consuming and non-alcohol consuming households within Bihar in (c) and (d). Food expenses include expenses on cereals, pulses, edible oils, ghee, spices, vegetables, potatoes, onions, milk, dairy products, bread, biscuits, mithai, namkeens, noodles, pasta, flakes, chocolates, cakes, ice-cream, jams, pickles, health supplements, eggs, meat, fish, ready-to-eat food, tea, coffee, beverages, bottled water, baby food and other food expenses. Clothing includes footwear, bags, wallets, watches, glasses, artificial jewellery, bindis, bangles, gems and jewellery. Regressions include district fixed effects and time (month $\times$ year) fixed effects and are weighted. Confidence intervals are at the 95 percent level and adjusted for state-level clustering in panel (a) and (b) and district-level clustering in panel (c) and (d). The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

Figure 6: Event Study Results: Alternative Expenses

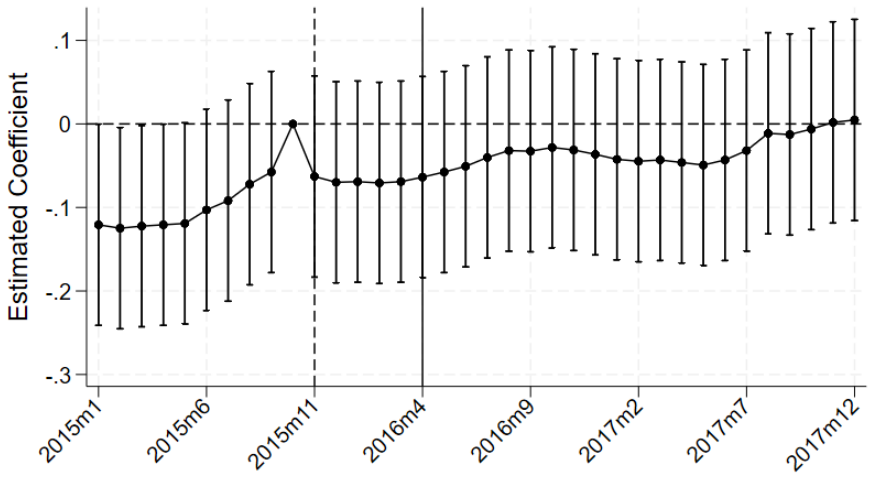


(a) Alcohol Consumption

(b) Cigarettes, Bidis, Tobacco

Notes: Each graph includes point estimates from the event study exercise (normalised to 0 in October 2015) comparing alcohol-consuming and non-alcohol-consuming households in Bihar, excluding those districts that share a border with the neighbouring states of Jharkhand, West Bengal, or Uttar Pradesh. This is done considering that those members who live close to the border of neighbouring states may end up having access to liquor by crossing the border, thereby biasing our estimates. Regressions include district fixed effects and time (month × year) fixed effects and are weighted. Confidence intervals are at the 95 percent level and adjusted for district-level clustering. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

Figure 7: Event Study Results within Bihar: Excluding border-sharing districts



Notes: Each graph includes point estimates from the event study exercise (normalised to 0 in October 2015) comparing Bihar prices for Pan, Tobacco, and Other Intoxicants (treated) to All India (control). Confidence intervals are at the 95 percent level. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively. This series comes from the Ministry of Statistics and Program Implementation, Government of India.

Figure 8: Event Study Results for Prices

Table 1: Summary Statistics

Variable (mean)	Bihar	Neighbouring States	All India	Source
Demographic Characteristics				
Population	104,099,452	324,076,590	1,210,854,977	Census, 2011
Age(%):				
0-18	47.83	42.47	38.98	Census, 2011
19-24	9.08	10.72	10.92	Census, 2011
25-64	38.17	41.25	44.24	Census, 2011
65+	4.51	4.97	5.46	Census, 2011
Family Size	4.393666	4.436544	4.179941	CPHS
Religion and Caste(%):				
Hindu	86.529	85.277	85.103	
SC	19.746	22.695	18.337	CPHS
ST	0.197	2.249	5.101	CPHS
Upper Caste	18.497	35.852	31.002	CPHS
OBC	48.071	24.477	31.103	CPHS
Muslim	13.471	13.861	9.767	CPHS
Region(%):				
Urban	66.087	68.531	70.446	CPHS
Rural	33.913	31.469	29.554	CPHS
Other Economic Indicators:				
Sex Ratio	918	936	940	Census, 2011
Literacy Rate (%)	61.80	70.11	74.04	Census, 2011
Average Monthly Household Consumption (in INR)				
Liquor	146.7215	81.69791	109.0791	CPHS
CBT	136.0652	219.0099	173.6987	CPHS
Health Expenses-1	41.83422	93.88968	68.55887	CPHS
Health Expenses-2	35.21682	57.59393	86.594	CPHS
Total Expense	6808.429	8471.364	9024.207	CPHS
Total Income	9599.63	11086.85	13177.98	CPHS
Sample Observations	75184	316110	1237746	

Notes: The means reported from CPHS are weighted and calculated based on pooled data over ten months, from January 2015 to October 2015, before the alcohol ban announcement was made. Here, the sex ratio is defined as the number of women per 1000 men; the literacy rate is defined by the percentage of the population aged 7 and above who can read and write. CBT is an acronym for cigarettes, bidis, and tobacco. SC, ST, and OBC are acronyms for Scheduled Caste, Scheduled Tribes, and Other Backward Castes, respectively. In our sample, 9138 unique households are surveyed in Bihar, out of which 5381 are alcohol consumers, whereas 3757 do not consume alcohol up to 10 months prior to the ban announcement. Similarly, 16887 alcohol-consuming households are unique in the neighbouring states. We winsorise the data to exclude outliers at 1% and 99%.

Table 2: Wild cluster bootstrap inference for event-study coefficients for alcohol consumption (inter-state)

Event time / test	Test statistic	Bootstrap p-value
Joint test (pre-treatment)	$F(9,3) = 9.7970$	0.3750
$t = 19$ (Announcement)	$t(3) = 1.5078$	0.3750
$t = 20$	$t(3) = -0.6011$	0.5000
$t = 21$	$t(3) = -2.4023$	0.3125
$t = 22$	$t(3) = -6.2966$	0.0000
$t = 23$	$t(3) = -17.9294$	0.0000
$t = 24$ (Implementation)	$t(3) = -14.6208$	0.0625
$t = 25$	$t(3) = -13.4767$	0.0000
$t = 26$	$t(3) = -10.4260$	0.0000
$t = 27$	$t(3) = -7.7123$	0.0000
$t = 28$	$t(3) = -7.8930$	0.0000
$t = 29$	$t(3) = -8.1887$	0.0000
$t = 30$	$t(3) = -8.1984$	0.0000
$t = 31$	$t(3) = -6.8025$	0.0000
$t = 32$	$t(3) = -6.5185$	0.0000
$t = 33$	$t(3) = -5.8532$	0.0000
$t = 34$	$t(3) = -5.6515$	0.0000
$t = 35$	$t(3) = -4.9965$	0.0000
$t = 36$	$t(3) = -5.9386$	0.0000
$t = 37$	$t(3) = -6.0568$	0.0625
$t = 38$	$t(3) = -5.3656$	0.0000
$t = 39$	$t(3) = -4.5284$	0.0000
$t = 40$	$t(3) = -4.0768$	0.0000
$t = 41$	$t(3) = -3.4930$	0.0000
$t = 42$	$t(3) = -3.5733$	0.0000
$t = 43$	$t(3) = -3.6215$	0.0000
$t = 44$	$t(3) = -3.2456$	0.0000

Notes: This table reports wild cluster bootstrap inference for event-study treatment effects from the specification with district fixed effects and state-clustered standard errors. Bootstrap p-values are based on 9,999 replications using Rademacher weights and clustering at the state level. The confidence intervals are missing because of the availability of only four states in the cluster. The omitted reference period is $t = 18$. The first row reports the joint test that all pre-treatment coefficients from $t = 9$ to $t = 17$ are equal to zero. Subsequent rows report period-specific bootstrap tests for post-treatment coefficients.

Table 3: Wild cluster bootstrap inference for event-study coefficients for CBT consumption (inter-state)

Event time / test	Test statistic	Bootstrap p-value
Joint test (pre-treatment)	$F(9, 3) = 36.8963$	0.0000
$t = 19$ (Announcement)	$t(3) = 0.2528$	0.5625
$t = 20$	$t(3) = -10.2354$	0.1875
$t = 21$	$t(3) = -9.2632$	0.0000
$t = 22$	$t(3) = -8.1195$	0.1250
$t = 23$	$t(3) = -8.0765$	0.1250
$t = 24$ (Implementation)	$t(3) = -6.4420$	0.1875
$t = 25$	$t(3) = -5.2120$	0.1250
$t = 26$	$t(3) = -4.6542$	0.1250
$t = 27$	$t(3) = -2.6409$	0.2500
$t = 28$	$t(3) = -1.9842$	0.2500
$t = 29$	$t(3) = -2.1163$	0.2500
$t = 30$	$t(3) = -4.0249$	0.1250
$t = 31$	$t(3) = -4.0371$	0.1250
$t = 32$	$t(3) = -3.1539$	0.1250
$t = 33$	$t(3) = -2.8057$	0.1875
$t = 34$	$t(3) = -2.7024$	0.1250
$t = 35$	$t(3) = -2.8276$	0.0000
$t = 36$	$t(3) = -3.1682$	0.0625
$t = 37$	$t(3) = -3.1781$	0.0000
$t = 38$	$t(3) = -1.8153$	0.2500
$t = 39$	$t(3) = -0.7041$	0.5000
$t = 40$	$t(3) = -0.5324$	0.6250
$t = 41$	$t(3) = 0.0468$	0.9375
$t = 42$	$t(3) = 0.2426$	0.7500
$t = 43$	$t(3) = 0.6653$	0.6875
$t = 44$	$t(3) = 0.7145$	0.6250

Notes: This table reports wild cluster bootstrap inference for event-study treatment effects from the specification with district fixed effects and state-clustered standard errors. Bootstrap p-values are based on 9,999 replications using Rademacher weights and clustering at the state level. The confidence intervals are missing because of the availability of only four states in the cluster. The omitted reference period is $t = 18$. The first row reports the joint test that all pre-treatment coefficients from $t = 9$ to $t = 17$ are equal to zero. Subsequent rows report period-specific bootstrap tests for post-treatment coefficients.

Table 4: Wild cluster bootstrap inference for within-Bihar alcohol event-study coefficients

Event time / test	Test statistic	Bootstrap p-value	95% confidence set
Joint test of pre-treatment	$F(9, 28) = 3.2642$	0.1984	–
$t = 19$ (Announcement)	$t(28) = -1.3510$	0.2132	$[-45.61,\ 8.446]$
$t = 20$	$t(28) = -2.3159$	0.0266	$[-106.1,\ -5.817]$
$t = 21$	$t(28) = -3.6837$	0.0016	$[-159.4,\ -38.52]$
$t = 22$	$t(28) = -4.8078$	0.0003	$[-198.2,\ -68.82]$
$t = 23$	$t(28) = -6.7848$	0.0003	$[-232.8,\ -114.0]$
$t = 24$ (Implementation)	$t(28) = -7.1248$	0.0003	$[-244.2,\ -124.5]$
$t = 25$	$t(28) = -7.2373$	0.0000	$[-254.1,\ -132.0]$
$t = 26$	$t(28) = -7.5727$	0.0003	$[-254.4,\ -137.6]$
$t = 27$	$t(28) = -7.6009$	0.0000	$[-278.1,\ -149.7]$
$t = 28$	$t(28) = -7.7861$	0.0001	$[-283.3,\ -155.5]$
$t = 29$	$t(28) = -7.5970$	0.0000	$[-285.1,\ -153.1]$
$t = 30$	$t(28) = -8.3472$	0.0000	$[-283.9,\ -160.9]$
$t = 31$	$t(28) = -8.3458$	0.0000	$[-285.5,\ -163.0]$
$t = 32$	$t(28) = -8.3522$	0.0000	$[-285.5,\ -164.2]$
$t = 33$	$t(28) = -8.4714$	0.0000	$[-287.0,\ -166.9]$
$t = 34$	$t(28) = -8.4893$	0.0000	$[-286.9,\ -165.7]$
$t = 35$	$t(28) = -8.4986$	0.0000	$[-286.8,\ -165.7]$
$t = 36$	$t(28) = -8.4934$	0.0000	$[-286.4,\ -165.6]$
$t = 37$	$t(28) = -8.5165$	0.0000	$[-286.1,\ -166.1]$
$t = 38$	$t(28) = -8.4437$	0.0000	$[-287.1,\ -165.2]$
$t = 39$	$t(28) = -8.4613$	0.0000	$[-285.6,\ -164.5]$
$t = 40$	$t(28) = -8.4587$	0.0000	$[-283.7,\ -163.6]$
$t = 41$	$t(28) = -8.4762$	0.0000	$[-283.3,\ -162.9]$
$t = 42$	$t(28) = -8.5060$	0.0000	$[-283.4,\ -164.5]$
$t = 43$	$t(28) = -8.4849$	0.0000	$[-284.7,\ -163.7]$
$t = 44$	$t(28) = -8.4962$	0.0000	$[-283.8,\ -164.4]$

Notes: This table reports wild cluster bootstrap inference for within-Bihar alcohol event-study treatment effects from the specification using manually constructed interaction terms (`int13`–`int44`). Bootstrap p-values are based on 9,999 replications using Rademacher weights and clustering at the district level. The omitted reference period is $t = 18$. The first row reports the joint test that all reported pre-treatment coefficients from $t = 13$ to $t = 17$ are equal to zero. Subsequent rows report period-specific bootstrap tests for post-treatment coefficients.

Table 5: Wild cluster bootstrap inference for within-Bihar CBT event-study coefficients

Event time / test	Test statistic	Bootstrap p-value	95% confidence set
Joint test of pre-treatment	$F(9, 28) = 1.1137$	0.6089	–
$t = 19$ (Announcement)	$t(28) = -0.4742$	0.6643	$[-22.75,\ 16.88]$
$t = 20$	$t(28) = -1.6942$	0.1498	$[-56.56,\ 11.38]$
$t = 21$	$t(28) = -2.0551$	0.1200	$[-71.45,\ 12.97]$
$t = 22$	$t(28) = -2.5944$	0.0587	$[-83.49,\ 2.034]$
$t = 23$	$t(28) = -3.0444$	0.0277	$[-107.7,\ -8.809]$
$t = 24$ (Implementation)	$t(28) = -2.8256$	0.0408	$[-104.7,\ -3.77]$
$t = 25$	$t(28) = -2.9728$	0.0295	$[-103.5,\ -6.674]$
$t = 26$	$t(28) = -2.6742$	0.0457	$[-108.7,\ -1.866]$
$t = 27$	$t(28) = -2.6333$	0.0425	$[-119.9,\ -3.644]$
$t = 28$	$t(28) = -3.0700$	0.0244	$[-128.3,\ -13.67]$
$t = 29$	$t(28) = -3.3375$	0.0200	$[-140.8,\ -16.39]$
$t = 30$	$t(28) = -3.4672$	0.0132	$[-149.5,\ -20.76]$
$t = 31$	$t(28) = -3.6777$	0.0110	$[-151.3,\ -27.06]$
$t = 32$	$t(28) = -3.8182$	0.0115	$[-157.6,\ -27.18]$
$t = 33$	$t(28) = -3.7158$	0.0140	$[-156.3,\ -23.51]$
$t = 34$	$t(28) = -3.7835$	0.0139	$[-157.5,\ -26.68]$
$t = 35$	$t(28) = -3.7465$	0.0116	$[-155.0,\ -28.23]$
$t = 36$	$t(28) = -3.8708$	0.0087	$[-156.3,\ -32.27]$
$t = 37$	$t(28) = -3.7359$	0.0119	$[-152.9,\ -28.48]$
$t = 38$	$t(28) = -2.7299$	0.0482	$[-146.4,\ -0.7529]$
$t = 39$	$t(28) = -2.3160$	0.0681	$[-147.8,\ 7.693]$
$t = 40$	$t(28) = -2.2105$	0.0785	$[-150.2,\ 12.04]$
$t = 41$	$t(28) = -1.9961$	0.0911	$[-139.1,\ 13.87]$
$t = 42$	$t(28) = -1.8221$	0.1156	$[-130.7,\ 18.33]$
$t = 43$	$t(28) = -1.6512$	0.1457	$[-110.0,\ 21.15]$
$t = 44$	$t(28) = -1.5088$	0.1804	$[-99.75,\ 23.67]$

Notes: This table reports wild cluster bootstrap inference for within-Bihar cigarettes, bidis, and tobacco (CBT) event-study treatment effects from the specification using manually constructed interaction terms (`int13`–`int44`). Bootstrap p-values are based on 9,999 replications using Rademacher weights and clustering at the district level. The omitted reference period is $t = 18$. The first row reports the joint test that all reported pre-treatment coefficients from $t = 13$ to $t = 17$ are equal to zero. Subsequent rows report period-specific bootstrap tests for post-treatment coefficients.

A Supplementary Analysis

A.1 Average Treatment Effects

We use the following empirical specification to estimate the average treatment effects following the ban announcement and its subsequent implementation.

$$Y_{it} = \beta_0 + \beta_1 \text{Treated}_i + \gamma_{\text{ann}} \text{Post}_1 \times \text{Treated}_i + \gamma_{\text{ban}} \text{Post}_2 \times \text{Treated}_i + \beta_4 X_i + \beta_5 X_{it} + \alpha_d + \alpha_t + \epsilon_{it} \quad (2)$$

where $Post_1$ is a dummy that indicates the post-announcement but pre-implementation period, where $Post_2$ represents an indicator that takes a value of 1 post the ban implementation. The results are listed below.

Table 6: Average Treatment Effect - Within Bihar

	(1)	(2)	(3)	(4)
	Alcohol	CBT	Health Expense 1	Health Expense 2
Treated	268.8***	32.52***	10.68***	-2.104***
	(345.46)	(31.11)	(6.80)	(-5.55)
post=1 × Treated	-154.6***	-15.11***	-17.12***	5.836***
	(-125.60)	(-9.13)	(-6.89)	(9.73)
post=2 × Treated	-269.9***	-48.57***	-32.95***	-6.018***
	(-314.58)	(-42.12)	(-19.02)	(-14.40)
Observations	259229	259229	259229	259229
R^2	0.574	0.219	0.061	0.272

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Average Treatment Effect (Bihar vs Neighboring States)

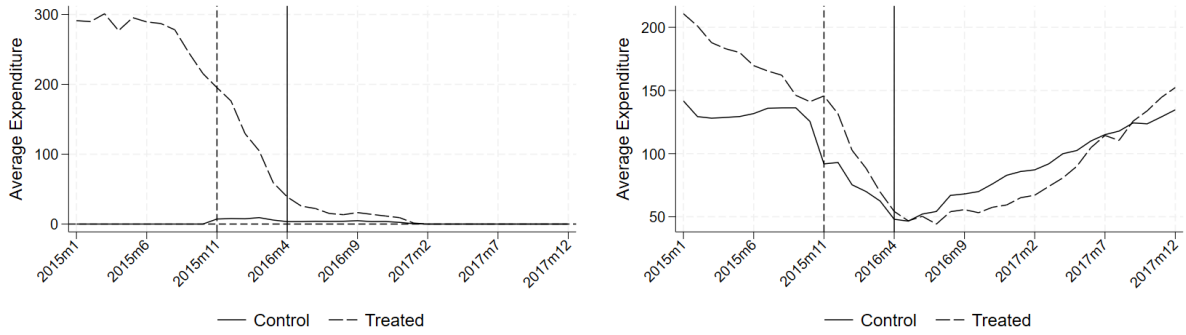
	(1) Alcohol	(2) CBT	(3) Health Expense 1	(4) Health Expense 2
post=1 × treated=1	-82.72***	-33.42*	28.36*	11.75*
	(-47.90)	(-3.46)	(3.43)	(4.94)
post=2 × treated=1	-153.9***	-40.35	-34.31	17.43
	(-16.12)	(-2.79)	(-1.90)	(2.24)
Observations	645324	645324	645324	645324
R^2	0.303	0.324	0.173	0.227

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

B Additional Figures

B.1 Raw trends for Within Bihar



(a) Liquor Expenses

(b) CBT Expenses

Notes: This series comes from the Consumer Pyramids Household Survey (CPHS). Panel (a) plots the average monthly expenses on liquor in the Indian state of Bihar for its alcohol-consuming (treated) and non-alcohol-consuming (control) households; Panel (b) plots the average monthly expenses on Cigarettes, Bidis, and other Tobacco (CBT) in Bihar for the same set of households. The dotted and bold vertical lines represent the months when the ban announcement was made (November 2015) and when it was implemented, April 2016, respectively.

Figure 9: Average Monthly Expenditure on Liquor and CBT within Bihar